MEDICAL INSURANCE COST PREDICITION USING MACHINE LEARNING

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INTRODUCTION

Accurate prediction of medical insurance costs is critical for ensuring fair pricing and effective financial planning in the healthcare sector. Traditional actuarial methods often fail to capture the complex, nonlinear relationships between influencing factors such as age, BMI, smoker status, and pre-existing health conditions. Machine learning (ML) offers a powerful alternative by leveraging large datasets to identify patterns and predict costs with higher precision. This project explores the application of ML algorithms, including Gradient Boosting and Random Forest, to develop a robust predictive model aimed at improving decision-making for insurers and policyholders alike

SCOPE

The scope of this project includes:

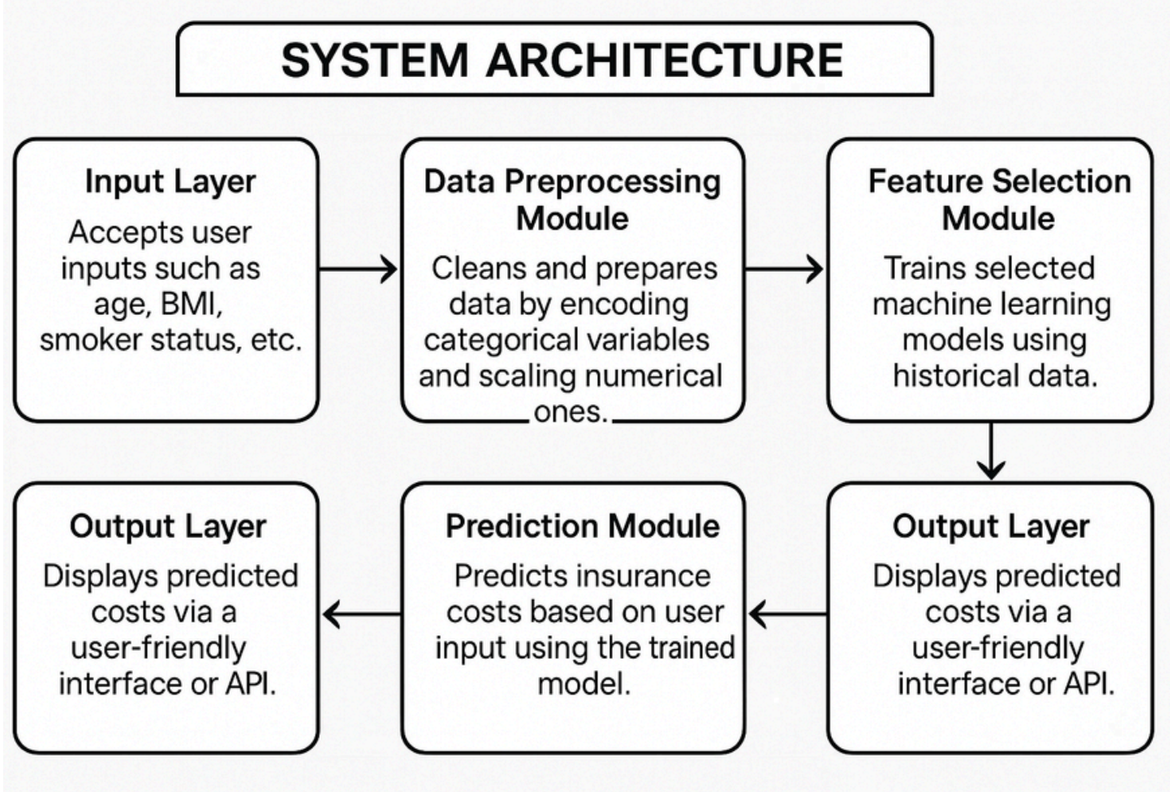
- Feature Analysis: Identifying key factors influencing insurance costs, such as age, BMI, smoker status, number of children, region, and medical history.
- Model Comparison: Evaluating multiple ML models (Gradient Boosting Regression, Random Forest Regression, Linear Regression) to identify the best-performing algorithm.
- Efficiency Improvement: Automating cost prediction to reduce manual effort and improve decision-making efficiency for insurers.
- Practical Application: Providing a scalable framework for predicting medical insurance costs that can be integrated into real-world systems for policy pricing and underwriting.

METHODOLOGY

The methodology involves the following steps:

1. Data Collection:
   - The dataset used is Kaggle's Medical Insurance Cost Dataset, containing features like age, BMI, smoker status, region, etc.
2. Data Preprocessing:
   - Handling missing values.
   - Encoding categorical variables (e.g., gender, region).
   - Scaling numerical features (e.g., BMI, age).
   - Removing outliers to ensure data quality.
3. Feature Selection:
   - Identifying influential features using statistical techniques or feature importance metrics.
4. Model Implementation:
   - Training multiple machine learning models: Gradient Boosting Regression (best performer), Random Forest Regression, Linear Regression.
   - Splitting the dataset into training (80%) and testing (20%) sets.
   - Hyperparameter tuning for optimal performance.
5. Evaluation Metrics:
   - Models are evaluated using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R2*R*2-score

SYSTEM ARCHITECTURE



RESULTS

```
   age     sex     bmi  children smoker     region      charges
0   19  female  27.900         0    yes  southwest  16884.92400
1   18    male  33.770         1     no  southeast   1725.55230
2   28    male  33.000         3     no  southeast   4449.46200
3   33    male  22.705         0     no  northwest  21984.47061
4   32    male  28.880         0     no  northwest   3866.85520
Missing Values:
 age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64

Model Performance:
Mean Absolute Error (MAE): 2500.28
Mean Squared Error (MSE): 20562168.67
R² Score: 0.8676
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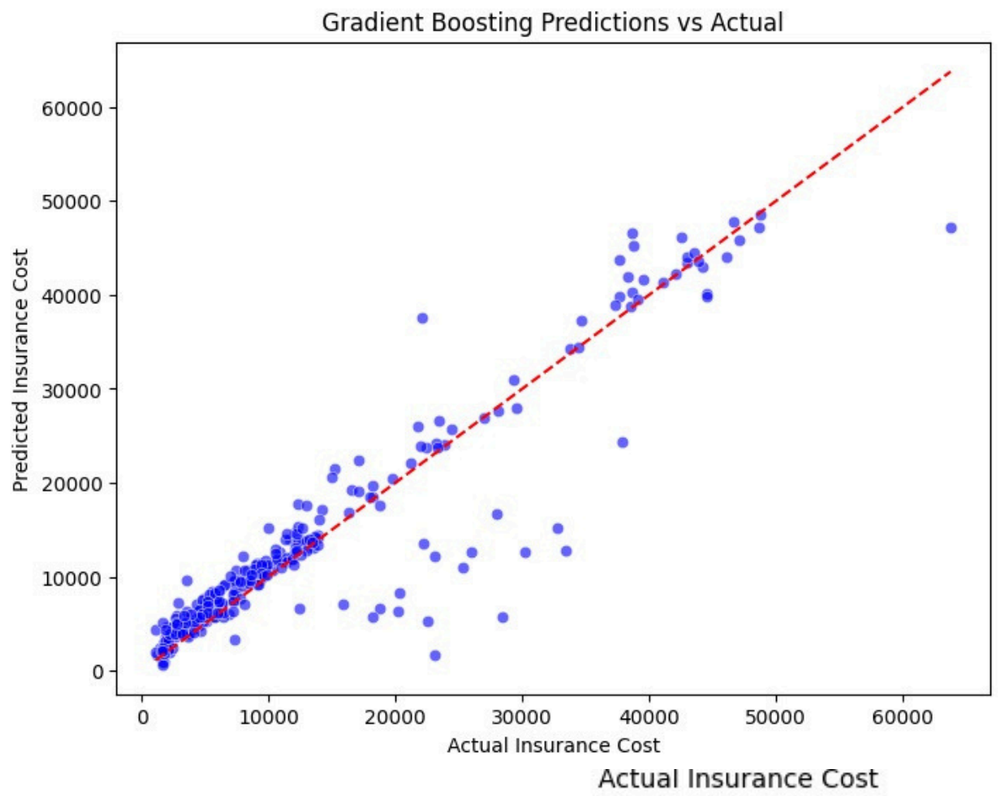

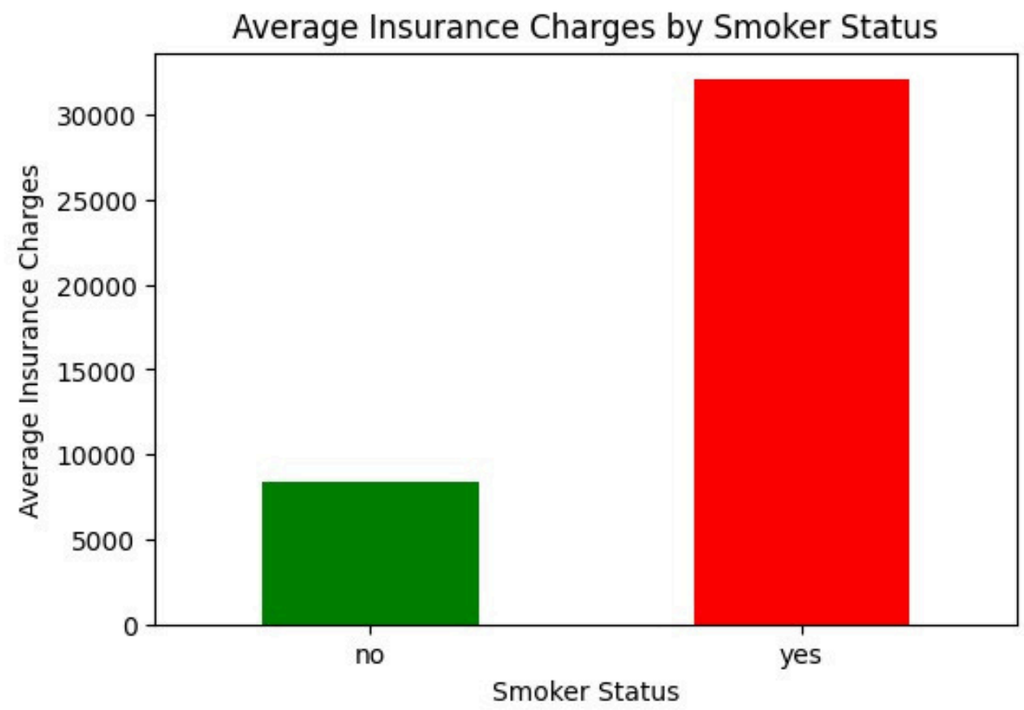

This project demonstrates the effectiveness of machine learning in predicting medical insurance costs, with the Gradient Boosting Regression model achieving 86.76% accuracy as shown by the R^2 score in the performance metrics. The analysis reveals smoking status as the most significant factor influencing insurance charges, with smokers paying nearly four times more than non-smokers as evidenced in the bar chart visualization. The scatter plot confirms the model's strong predictive capability, with most predictions closely following the ideal prediction line despite some variance at higher cost ranges. This automated framework not only reduces manual effort in cost estimation but also enhances transparency in policy pricing, ultimately enabling insurers to optimize resource allocation and helping policyholders plan their finances more effectively through data-driven methodologies.

CONCLUSION

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